

Yinghao (Peter) Guan

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Education

Santa Monica College , Computer Science	Sept 2025 – Expected Transfer: Fall 2027
University of Edinburgh , Studied Computer Science Relevant Coursework: Software Engineering, Data Structure, Reasoning and Agents	Sept 2022 – July 2024
University of Auckland , Studied Engineering & Finance.	Feb 2022 – May 2022

Skills

- **Languages:** Python, Java, C#, R, JavaScript, HTML/CSS, C++, Solidity
- **Frameworks & Libraries:** Pandas, Scikit-learn, Matplotlib, JavaFX, JUnit, Unity Engine, ANTLR4, rpy2
- **Databases & Platforms:** Firebase, Linux, Git, Maven, WSL
- **Knowledge Areas:** Explainable AI (XAI), HCI, Data Structures & Algorithms, OOP, MVC/MVVM

Hackathon Experience

AdaHack, AI Language-Learning Game	Spring 2024
• Designed and built a functional prototype for an AI-powered language learning game using Unity and C#. Engineered core gameplay mechanics and AI interaction within a 48-hour timeframe. Submitted the project to Devpost.	
Encode ETH London, DeFi Application Prototype	Fall 2023
• Collaborated in a team to develop a Web3 dApp for decentralized finance. Wrote and tested Solidity smart contracts and integrated them with a front-end.	

Projects

Custom Markup Language Editor & Converter	Aug 2025 - Present
• Engineered a desktop application with JavaFX and ANTLR4 to parse a custom markup language into an Abstract Syntax Tree (AST). • Designed a robust system using the Visitor pattern to convert the AST to a Pandoc AST, enabling seamless conversion to HTML and LaTeX. • Implemented the application using the MVVM architecture, enhancing its extensibility and testability.	
UCLA HCI Research Collaboration	Jul 2025 – Present
• Led a research project with a UCLA HCI faculty member, investigating the impact of eXplainable AI (XAI) on user trust and empathy in emotional support LLMs. • Designed and conducted an A/B test, implementing two distinct AI models – a baseline LLM and a version integrated with XAI and sentiment analysis for user emotion detection. • Analysed user survey data and model outputs to perform both qualitative and quantitative analysis, preparing to publish a peer-reviewed paper as the sole author.	
Independent Research: Behavioural Modelling of Excessive Trading	Jan 2025 – Apr 2025
• Architected the technical pipeline for a team research project, investigating the effects of behavioural biases (FoMO, Anchoring, Herding) on excessive trading. • Developed a comprehensive data analysis workflow using R for the core PLS-SEM modelling, and Python (pandas) for data acquisition and cleaning. • Engineered a hybrid computational environment in WSL to seamlessly integrate Python and R (via rpy2) for complex data processing and statistical analysis.	